# AGRISENSE AI

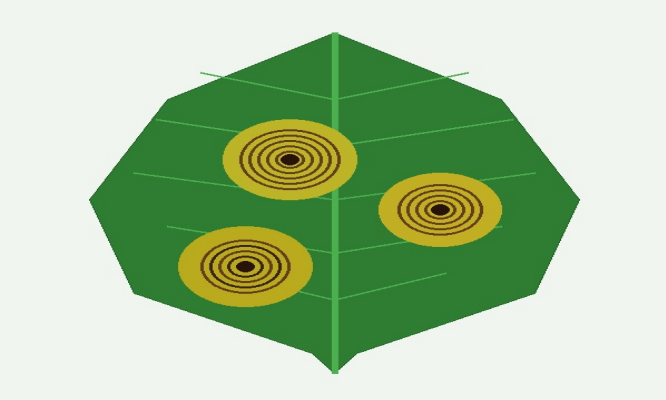
**Intelligent Agriculture Decision Support Platform**

**DIAGNOSTIC REPORT**
**Report ID:** 5F43E916
**Date:** 18 Aug 2026, 04:01 PM

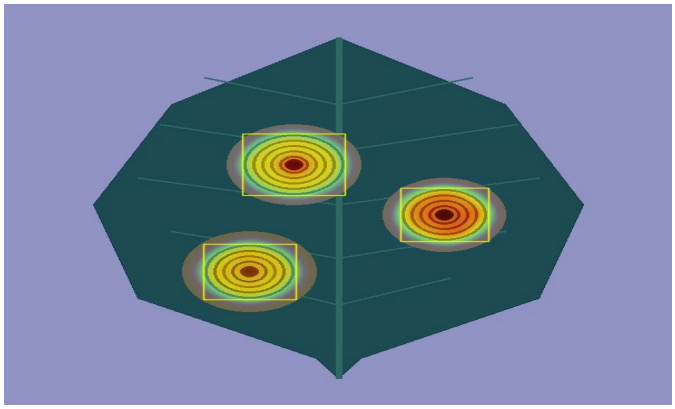
**Farmer:** Ramesh Gowda | **Location:** Mysuru, Karnataka | **Crop Analyzed:** Tomato

| | |
|---|---|
| **DIAGNOSED CONDITION:** | **Early Blight *(Alternaria solani)*** |
| **Confidence Level:** | **95.7%** (EfficientNetB3 Saliency) |
| **Severity Rating:** | **MODERATE** (Estimated affected canopy: **14.2%**) |
| **Pathogen Class:** | Fungal |

**Original Leaf Specimen**



**Explainable Grad-CAM Heatmap Attention**



## Clinical Symptoms & Biological Cause

**Symptoms:** Dark brown to black spots with characteristic concentric rings ('bullseye' pattern) surrounded by yellow chlorotic halos, starting on lower older leaves.

**Etiology/Cause:** Fungus Alternaria solani thriving in warm, humid conditions with alternate wet and dry periods.

## Recommended Treatment Protocol

| Treatment Type | Prescription & Application Details |
|---|---|
| **Organic / Bio-Control** | Spray Neem oil (5ml/L) or Copper oxychloride (2.5g/L). Apply Trichoderma viride bio-fungicide (5g/L) to soil and foliage. Prune infected bottom leaves and avoid overhead splashing. |
| **Chemical / Fungicide** | Foliar spray of Mancozeb 75% WP @ 2g/L or Chlorothalonil 75% WP @ 2g/L at 7-10 day intervals. In severe outbreaks, use Azoxystrobin 23% SC @ 1ml/L.<br>**Dosage:** Mancozeb 2g per liter water (500g/acre in 200L water).<br>**Frequency:** Every 7 to 10 days until new foliage is symptom-free. |
| **Preventive Measures** | Implement 3-year crop rotation (avoid Solanaceous crops), use drip irrigation, mulch soil surface to prevent spore splash, and maintain proper plant spacing for aeration. |

*Note: This clinical report was generated by AgriSense AI Diagnostic Engine using deep learning visual analysis. For epidemic outbreaks or severe crop stress, please consult your nearest Krishi Vigyan Kendra (KVK) or Regional Agriculture Extension Officer.*